

DETERMINANTS

LEVEL-VI

SINGLE ANSWER QUESTIONS

1. If
$$\begin{vmatrix} \operatorname{cosec} \alpha & 1 & 0 \\ 1 & 2\operatorname{cosec} \alpha & 1 \\ 0 & 1 & 2\operatorname{cosec} \alpha \end{vmatrix} = \frac{1}{2} \left(\tan^3 \frac{\alpha}{\lambda} + \cot^3 \frac{\alpha}{\mu} \right)$$
 then the value of

$\lambda + \mu$ is

- A) 5 B) 2
C) 3 D) 4

2. If $ax_1^2 + by_1^2 + cz_1^2 = ax_2^2 + by_2^2 + cz_2^2 = ax_3^2 + by_3^2 + cz_3^2 = \lambda$
and $ax_1x_2 + by_1y_2 + cz_1z_2 =$
 $ax_2x_3 + by_2y_3 + cz_2z_3 = ax_3x_1 + by_3y_1 + cz_3z_1 =$

$= \mu$ &
$$\begin{vmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix} = (\lambda - \mu) \left(\frac{\lambda + \theta\mu}{abc} \right)^{\frac{1}{\theta}}$$

then the value of $\theta + \beta$ is

- A) 4 B) 3 C) 2 D) 1

3. If $\Delta(x) = \begin{vmatrix} \alpha + x & \theta + x & \lambda + x \\ \beta + x & \phi + x & \mu + x \\ \gamma + x & \psi + x & \nu + x \end{vmatrix}$ then find

the value of $\Delta''(x)$

- A) $\alpha + \beta + \gamma$ B) 0
C) $2\alpha + \beta$ D) 3β

4. Let
$$\Delta_1 = \begin{vmatrix} \log \left(\frac{a}{b} \right) & \log \left(\frac{b}{c} \right) & \log \left(\frac{c}{a} \right) \\ \log \left(\frac{b}{c} \right) & \log \left(\frac{c}{a} \right) & \log \left(\frac{a}{b} \right) \\ \log \left(\frac{c}{a} \right) & \log \left(\frac{a}{b} \right) & \log \left(\frac{b}{c} \right) \end{vmatrix}$$

and $\Delta_2 = \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}, [a \neq b \neq c]$ then the

value of $\frac{\Delta_1}{\Delta_2}$ is

- A) $\frac{1}{4}(a-b)(b-c)(c-a)$
B) $-(a+b+c)$
C) 0
D) $\log(a+b+c)$

5. If

$$D_k = \begin{vmatrix} 3^k & \frac{1}{(k+1)(k+2)} & \cos(k+1)\theta \\ \frac{3^n-1}{2} & \frac{n}{n+1} & \frac{\sin \frac{n\theta}{2} \cos \frac{(n-1)\theta}{2}}{\sin \frac{\theta}{2}} \\ a & b & c \end{vmatrix}$$

then $\sum_{k=0}^{n-1} D_k$ is

- A) independent of n
B) independent of a, b, c
C) $a + b + c$ D) $a + 2b + c$

6. If the value of a fourth order determinant is 3, then the value of the determinant formed by the cofactors is

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- A) 27 B) 64 C) 1 D) 81

7. In ΔABC , it is being given that

$$\Delta = \begin{vmatrix} 1 & 1 & 1 \\ \cot \frac{A}{2} & \cot \frac{B}{2} & \cot \frac{C}{2} \\ \tan \frac{B}{2} + \tan \frac{C}{2} & \tan \frac{A}{2} + \tan \frac{C}{2} & \tan \frac{A}{2} + \tan \frac{B}{2} \end{vmatrix} = 0$$

then triangle must be

- A) equilateral B) isosceles
C) obtuse angled D) right angled

8. The value of the determinant

$$\Delta = \begin{vmatrix} 1^2 & 2^2 & 3^2 & 4^2 \\ 2^2 & 3^2 & 4^2 & 5^2 \\ 3^2 & 4^2 & 5^2 & 6^2 \\ 4^2 & 5^2 & 6^2 & 7^2 \end{vmatrix} \text{ is equal to}$$

- A) 1 B) 0 C) 2 D) 3

9. If

$$p\lambda^4 + p\lambda^3 + r\lambda^2 + s\lambda + t = \begin{vmatrix} \lambda^2 + 3\lambda & \lambda^2 + 1 & \lambda + 3 \\ \lambda^2 + 1 & 2 - \lambda & \lambda + 3 - 3 \\ \lambda^2 - 3 & \lambda + 4 & 3\lambda \end{vmatrix}$$

then p is equal to

- A) -5 B) -4 C) -3 D) -2

$$10. \text{ If } \begin{vmatrix} a^2 + \lambda^2 & ab + c\lambda & ca - b\lambda \\ ab - c\lambda & b^2 + \lambda^2 & bc - a\lambda \\ ca - b\lambda & bc - a\lambda & c^2 + \lambda^2 \end{vmatrix} = \begin{vmatrix} \lambda & c & -b \\ -c & \lambda & a \\ b & b & \lambda \end{vmatrix}$$

$= (1 + a^2 + b^2 + c^2)^3$ then the value of λ is

- A) 8 B) 27 C) 1 D) -1

11. If $c < 1$ and the system of equations $x + y - 1 = 0$, $2x - y - c = 0$ and $bx + 3by - c = 0$ is consistent then the possible real values of b are

- A) $b \in \left(-3, \frac{3}{4}\right)$ B) $b \in \left(\frac{3}{2}, 4\right)$

- C) $b \in \left(-\frac{3}{4}, 3\right)$ D) $b \in \left(-3, \frac{3}{2}\right)$

$$12. \text{ If } g(x) = \begin{vmatrix} a^{-x} & e^{x \log_e a} & x^2 \\ a^{-3x} & e^{3x \log_e a} & x^4 \\ a^{-5x} & e^{5x \log_e a} & 1 \end{vmatrix} \text{ then}$$

- A) graphs of $g(x)$ is symmetrical about origin
B) graphs of $g(x)$ is symmetrical about y-axis

$$C) \frac{d^4 g(x)}{dx^4} \Big|_{x=0} \neq 0$$

$$D) f(x) = g(x) \log \left(\frac{a-x}{a+x} \right) \text{ is an odd function}$$

$$13. \begin{vmatrix} yz - x^2 & zx - y^2 & xy - z^2 \\ xz - y^2 & xy - z^2 & yz - x^2 \\ xy - z^2 & yz - x^2 & zx - y^2 \end{vmatrix} = \begin{vmatrix} r^2 & u^2 & u^2 \\ u^2 & r^2 & u^2 \\ u^2 & u^2 & r^2 \end{vmatrix}$$

then

- A) $r^2 = x + y + z$ B) $r = x^2 + y^2 + z^2$
C) $u^2 = yz + zx + xy$ D) $u^2 = xyz$

$$14. \begin{vmatrix} my + nz & mq + nr & mb + nc \\ kz - mx & kr - mp & kc - ma \\ -nx - ky & -np - kq & -na - kb \end{vmatrix} \text{ is equal to}$$

- A) 0 B) -1 C) 1 D) 2

15. Consider the system of equations $x - 2y + 3z = -1$, $x - 3y + 4z = 1$ and $-x + y - 2z = k$

Statement - I : The system of equations has no solution for $k \neq 3$

Statement - II : The determinant

$$\begin{vmatrix} 1 & 3 & -1 \\ -1 & -2 & k \\ 1 & 4 & 1 \end{vmatrix} \neq 0 \text{ for } k \neq 3$$

[IITJEE 2008]

A) Statement - I is true, Statement - II is true, Statement - II is a correct explanation for Statement - I

B) Statement - I is true, Statement - II is true, Statement - II is not a correct explanation for Statement - I

C) Statement - I is true, Statement - II false

D) Statement - I is false, Statement - II true

MULTIPLE ANSWER QUESTIONS

16. If $\Delta = \begin{vmatrix} x^2 - a^2 & x^2 - b^2 & x^2 - c^2 \\ (x-a)^3 & (x-b)^3 & (x-c)^3 \\ (x+a)^3 & (x+b)^3 & (x+c)^3 \end{vmatrix} = 0$

(where $a \neq b \neq c$) then value of x is/are

A) 0 B) $\sqrt{\frac{1}{3}(bc+ca+ab)}$

C) $-\sqrt{\frac{1}{3}(bc+ca+ab)}$ D) a

17. The determinant

$$\Delta = \begin{vmatrix} a^2 + x^2 & ab & ac \\ ab & b^2 + x^2 & bc \\ ac & bc & c^2 + x^2 \end{vmatrix}$$

is divisible by

A) x B) x^2 C) x^3 D) x^4

18. Let a, b, c are non-zero real number and

$$\Delta = \begin{vmatrix} bc & ca & ab \\ ca & ab & bc \\ ab & bc & ca \end{vmatrix}$$

where (ω) is cube root of unity, then

Δ vanishes iff

A) $\frac{1}{a} + \frac{1}{b\omega} + \frac{1}{c\omega^2} = 0$ B) $\frac{1}{a} + \frac{1}{b\omega^2} + \frac{1}{c\omega} = 0$

C) $\frac{1}{a\omega} + \frac{1}{b\omega^2} + \frac{1}{c\omega} = 0$ D) $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = 0$

MATRIX MATCHING QUESTIONS

19. The entries in a 3×3 determinant are either 1 or -1, then

Column - I

Column - II

A) Total number of such determinants

p) 4

B) The sum of maximum & minimum trace of the determinants possible

q) 3

C) The maximum value of such a determinant

r) 512

D) The maximum value of trace of such determinant

s) zero

20. Match the following

Column - I

Column - II

A) $f(x) = \begin{vmatrix} \cos x & x & 1 \\ 2\sin x & x^2 & 2x \\ \tan x & x & 1 \end{vmatrix}$

p) π

then $\lim_{x \rightarrow 0} \frac{f(x)}{x}$

B) $f(x) = \begin{vmatrix} 2\cos^2 2x & \sin 2x & -\sin x \\ \sin 2x & 2\sin^2 x & \cos x \\ \sin x & -\cos x & 0 \end{vmatrix}$

q) 2

then $\int_0^{\pi/2} f'(x) dx =$

C) If $\Delta = \begin{vmatrix} f(x) & f(x) + f\left(\frac{1}{x}\right) \\ 1 & f\left(\frac{1}{x}\right) \end{vmatrix} = 0$ r) 0

where it is given that

$f(x) = a + bx^n$ and

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$$f(2) = 17, \text{ then } f(5) = 313k$$

then $k =$

$$D) \text{ If } \Delta = \begin{vmatrix} 2\cos^2 x & \sin 2x & -\sin x \\ \sin 2x & 2\sin^2 x & \cos x \\ \sin x & -\cos x & 0 \end{vmatrix} \text{ s) } 1$$

$$\text{then } \int_0^{\pi/2} (\Delta + \Delta') dx = k\pi$$

then the value of k is

COMPREHENSION QUESTIONS

Passage - I

If

$$D = \begin{vmatrix} x^2 + x + 2 & 2x^2 + 3x + 1 & 3x^2 + 5x + 3 \\ 20x^2 + 20x + 59 & 40x^2 + 60x + 20 & 60x^2 + 100x + 70 \\ 2x^2 + 2x + 6 & 4x^2 + 6x + 2 & 6x^2 + 10x + 7 \end{vmatrix}$$

$= ax^2 + bx + c$, then

21. The value of 'a' is

$$A) \begin{vmatrix} 1 & 2 & 3 \\ 1 & 0 & 1 \\ 2 & 1 & 1 \end{vmatrix} \quad B) \begin{vmatrix} 1 & 2 & 3 \\ 1 & 1 & 1 \\ 2 & 0 & 1 \end{vmatrix}$$

$$C) \begin{vmatrix} 1 & 2 & 3 \\ 1 & 0 & 1 \\ 2 & 0 & 1 \end{vmatrix} \quad D) \begin{vmatrix} 1 & 2 & 3 \\ 1 & 0 & 1 \\ 0 & 2 & 1 \end{vmatrix}$$

22. The value of 'b' is

$$A) \begin{vmatrix} 1 & 3 & 5 \\ 1 & 0 & 1 \\ 2 & 0 & 1 \end{vmatrix} \quad B) \begin{vmatrix} 1 & 3 & 5 \\ 1 & 0 & 1 \\ 2 & 1 & 1 \end{vmatrix}$$

$$C) \begin{vmatrix} 1 & 3 & 5 \\ 2 & 0 & 1 \\ 1 & 1 & 1 \end{vmatrix} \quad D) \begin{vmatrix} 1 & 3 & 5 \\ 1 & 1 & 1 \\ 2 & 0 & 1 \end{vmatrix}$$

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23. The value of 'c' is

$$A) \begin{vmatrix} 2 & 1 & 3 \\ 1 & 0 & 1 \\ 2 & 1 & 1 \end{vmatrix} \quad B) \begin{vmatrix} 2 & 1 & 3 \\ 1 & 0 & 1 \\ 2 & 0 & 1 \end{vmatrix}$$

$$C) \begin{vmatrix} 2 & 1 & 1 \\ 3 & 2 & 0 \\ 1 & 1 & 1 \end{vmatrix} \quad D) \begin{vmatrix} 2 & 1 & 3 \\ 1 & 0 & 1 \\ 2 & 1 & 0 \end{vmatrix}$$

Passage - II

Let p be an odd prime number and T_p be the following set of 2×2 matrices

$$T_p = \left\{ A = \begin{bmatrix} a & b \\ c & a \end{bmatrix}; a, b, c \in \{0, 1, 2, \dots, (p-1)\} \right\}$$

24. The number of A in T_p such that A is either symmetric or skew-symmetric or both, and $\det(A)$ is divisible by p is

$$A) (p-1)^2 \quad B) 2(p-1)$$

$$C) (p-1)^2 + 1 \quad D) 2p - 1$$

25. The number of A in T_p such that the trace of A is not divisible by p but $\det(A)$ is divisible by p is

$$A) (p-1)(p^2 - p + 1) \quad B) p^3 - (p-1)^2$$

$$C) (p-1)^2 \quad D) (p-1)(p^2 - 2)$$

26. The number of A in T_p such that $\det(A)$ is not divisible by p is

$$A) 2p^2 \quad B) p^3 - 5p$$

$$C) p^3 - 5p \quad D) p^3 - p^2$$

Passage - III

$$\text{If } a \neq b \neq c \text{ and } \begin{vmatrix} a & a^3 & a^4 - 1 \\ b & b^3 & b^4 - 1 \\ c & c^3 & c^4 - 1 \end{vmatrix} = 0 \text{ then}$$

27. If $2ab^2c = 3 - a^2c^2$ then a, b, c are in
(if $ac \neq \pm 1$)

$$A) AP \quad B) GP \quad C) HP \quad D) AGP$$

28. If $b^4 = 1$ then a,b,c are in
A) AP B) GP C) HP D) AGP
29. If $(a+c) = b(3a^2c^2 - 1)$ then a,b,c are in
A) AP B) GP C) HP D) AGP

Passage - IV

Let α, β be the roots of $ax^2 + bx + c = 0$
and let $S_n = \alpha^n + \beta^n$, for $n \geq 1$ and

$$\Delta = \begin{vmatrix} 3 & 1+S_1 & 1+S_2 \\ 1+S_1 & 1+S_2 & 1+S_3 \\ 1+S_2 & 1+S_3 & 1+S_4 \end{vmatrix}$$

30. If $\Delta_1 = \begin{vmatrix} 1 & 1 & 1 \\ 1 & \alpha & \beta \\ 1 & \alpha^2 & \beta^2 \end{vmatrix}$ then $\frac{\Delta}{\Delta_1^2}$ is equal to

A) 1 B) 2 C) 0 D) 3

31. Value of Δ in terms of α, β is

- A) $((\alpha+1)(\alpha-\beta)(\beta-1))^2$
B) $((\alpha-1)(\alpha-\beta)(\beta+1))^2$
C) $((\alpha-1)(\alpha+\beta)(\beta-1))^2$
D) $((\alpha-1)(\alpha-\beta)(\beta-1))^2$

32. Value of Δ in terms of a,b,c is

- A) $\frac{(a+b+c)^2(b^2-4ac)}{a^4}$
B) $\frac{(a+b+c)^2(b^2+4ac)}{a^4}$
C) $\frac{(a+b-c)^2(b^2+4ac)}{a^4}$
D) $\frac{(a+b+c)^2(b^2-4ac)}{b^4}$

Passage - V

Given that the system of equations $x=cy+bz$,
 $y=az+cx$, $z=bx+ay$ has non-zero solutions and
at least one of the a, b, c is a proper fraction.

33. $a^2 + b^2 + c^2$ is
A) >2 B) >3 C) <3 D) <2
34. abc is
A) >-1 B) >1 C) <2 D) <3
35. system has solution such that
A) $x:y:z = (1-2a^2):(1-2b^2):(1-2c^2)$
B) $x:y:z = \frac{1}{1-2a^2}:\frac{1}{1-2b^2}:\frac{1}{1-2c^2}$
C) $x:y:z = \frac{a}{1-a^2}:\frac{b}{1-b^2}:\frac{c}{1-c^2}$
D) $x:y:z = \sqrt{1-a^2}:\sqrt{1-b^2}:\sqrt{1-c^2}$

INTEGER ANSWER QUESTIONS

36. If $\begin{vmatrix} \frac{a^2+b^2}{c} & c & c \\ a & \frac{b^2+c^2}{a} & a \\ b & b & \frac{c^2+a^2}{b} \end{vmatrix} = kabc$

Where k is a numerical quantity then k should be

37. The system $ax + y + z = a$, $x + by + z = b$
and $x + y + cz = c$ has no solution if a,b,c
are all different from the number -----
38. The value of θ for which the system
 $(\sin 3\theta)x - y + z = 0$, $(\cos 2\theta)x + 4y + 3z = 0$,
 $2x + 7y + 7z = 0$ has non-trivial solution is
 $\frac{5\pi}{\lambda}$, where λ is a positive integer greater
than 1 then λ must be equal to

39. If $D_i = \begin{vmatrix} 1 & n & n \\ 2i & n^2 + n + 1 & n^2 + n \\ 2i-1 & n^2 & n^2 + n + 1 \end{vmatrix}$ and
 $\sum_{i=1}^n D_i = 56$ then n must be

40. If $\Delta(x) = \begin{vmatrix} 0 & 2x-2 & 2x+8 \\ x-1 & 4 & x^2+7 \\ 0 & 0 & x+4 \end{vmatrix}$ and

$$f(x) = \sum_{j=1}^3 \sum_{i=1}^3 a_{ij} c_{ij}, \text{ where } a_{ij} \text{ is the}$$

element of i^{th} row and j^{th} column in $\Delta(x)$ and c_{ij} is the cofactor of a_{ij} , for all ij then greatest value of $f(x)$, for $x \in [-3, 18]$ is

41. If A, B, C are angles of a triangle and $\sum \tan A \tan B = 9$, $x = \tan A$, $y = \tan B$, $z = \tan C$ then the numerical value of

$$\begin{vmatrix} x & y & z \\ y & z & x \\ z & x & y \end{vmatrix} \text{ is}$$

42. Number of positive integral solution (x, y, z) of the equation

$$\begin{vmatrix} x^3+1 & x^2y & x^2z \\ xy^2 & y^3+1 & y^2z \\ xz^2 & yz^2 & z^3+1 \end{vmatrix} = 11 \text{ is}$$

43. Given $(1+x+x^2)^8 = \sum_{k=0}^{16} a_k x^k$ then the

value of $\begin{vmatrix} a_3 & a_7 & a_{13} \\ a_4 & a_6 & a_{12} \\ a_5 & a_4 & a_{11} \end{vmatrix}$ is

44. If $a_1, a_2, a_3, 5, 4, a_7, a_8, a_9$ are in H.P and

$$D = \begin{vmatrix} a_1 & a_2 & a_3 \\ 5 & 4 & a_6 \\ a_7 & a_8 & a_9 \end{vmatrix} \text{ then the value of } [D] \text{ is}$$

(where $[]$ represents G.L.F)

45. If $a_1, a_2, a_3, \dots, a_{12}$ are in A.P and

$$\Delta_1 = \begin{vmatrix} a_1 a_5 & a_1 & a_2 \\ a_2 a_6 & a_2 & a_3 \\ a_3 a_7 & a_3 & a_4 \end{vmatrix} \Delta_2 = \begin{vmatrix} a_2 a_{10} & a_2 & a_3 \\ a_3 a_{11} & a_3 & a_4 \\ a_4 a_{12} & a_4 & a_5 \end{vmatrix}$$

then $\Delta_1 : \Delta_2 =$

46. Let $P = [a_{ij}]$ be 3×3 matrix and Let $Q = [b_{ij}]$ where $b_{ij} = 2^{i+j} a_{ij}$ for $1 \leq i, j \leq 3$. If the determinant of P is 2 and the determinant of the matrix Q is k^{13} then k is

47. Let ω be the complex number $\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}$. Then the number of distinct complex numbers Z satisfying

$$\begin{vmatrix} z+1 & \omega & \omega^2 \\ \omega & z+\omega^2 & 1 \\ \omega^2 & 1 & z+\omega \end{vmatrix} = 0 \text{ is}$$

[IITJEE 2010]

48. Let k be a positive real number and let

$$A = \begin{vmatrix} 2k-1 & 2\sqrt{k} & 2\sqrt{k} \\ 2\sqrt{k} & 1 & -2k \\ -2\sqrt{k} & 2k & -1 \end{vmatrix} \text{ and } B = \begin{vmatrix} 0 & 2k-1 & \sqrt{k} \\ 1-2k & 0 & 2\sqrt{k} \\ -\sqrt{k} & -2\sqrt{k} & 0 \end{vmatrix}$$

If $\det(\text{adj } A) + \det(\text{adj } B) = 10^6$ then $[k]$ is equal to [Note : $\text{adj } M$ denotes the adjoint of a square matrix M and $[k]$ denotes the largest integer less than or equal to k]

[IITJEE 2010]

SUBJECTIVE QUESTIONS

49. Show that

$$\begin{vmatrix} 1 & (m+n-l-p)^2 & (m+n-l-p)^4 \\ 1 & (n+l-m-p)^2 & (n+l-m-p)^4 \\ 1 & (l+m-n-p)^2 & (l+m-n-p)^4 \end{vmatrix} =$$

$$64(l-m)(l-n)(l-p)(m-n)(m-p)(n-p)$$

50. If $a = \cos \theta + i \sin \theta$, $b = \cos 2\theta - i \sin 2\theta$ and $c = \cos 3\theta + i \sin 3\theta$ and if

$$\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = \theta \text{ then } \theta = 2n\pi, n \in \mathbb{N}$$

51. If $f(x) = \begin{vmatrix} x^2 + x & 2x - 1 & x + 3 \\ 3x + 1 & 2 + x^2 & x^3 - 3 \\ x - 3 & x^2 + 4 & 2x \end{vmatrix}$

$$= a_0x^7 + a_1x^6 + a_2x^5 + \dots + a_6x + a_7$$

then prove that $(a_6, a_7) = (75, 21)$

52. If $\alpha_1\gamma_1 + \alpha_2\gamma_2 + \alpha_3\gamma_3$

$$= \beta_1\gamma_1 + \beta_2\gamma_2 + \beta_3\gamma_3 = 0$$

$$\gamma_1^2 + \gamma_2^2 + \gamma_3^2 = 1 \text{ and } \alpha_1\beta_1 + \alpha_2\beta_2 + \alpha_3\beta_3$$

$$= \frac{\sqrt{3}}{2} \sqrt{(\alpha_1^2 + \alpha_2^2 + \alpha_3^2)(\beta_1^2 + \beta_2^2 + \beta_3^2)}$$

Then show that

$$\begin{vmatrix} \alpha_1 & \alpha_2 & \alpha_3 \\ \beta_1 & \beta_2 & \beta_3 \\ \gamma_1 & \gamma_2 & \gamma_3 \end{vmatrix} = \frac{1}{4} (\alpha_1^2 + \alpha_2^2 + \alpha_3^2) (\beta_1^2 + \beta_2^2 + \beta_3^2)$$

53. If u, v, w be functions of x and if suffixes denote differentiation with respect to x

prove that $\frac{d}{dx} \begin{vmatrix} u_1 & v_1 & w_1 \\ u_2 & v_2 & w_2 \\ u_3 & v_3 & w_3 \end{vmatrix} = \begin{vmatrix} u_1 & v_1 & w_1 \\ u_2 & v_2 & w_2 \\ u_4 & v_4 & w_4 \end{vmatrix}$

54. If f, g and h are differentiable functions of x

and $\Delta = \begin{vmatrix} f & g & h \\ (xf)' & (xg)' & (xh)' \\ (x^2f)'' & (x^2g)'' & (x^2h)'' \end{vmatrix}$ then

prove that $\Delta = \begin{vmatrix} f & g & h \\ f' & g' & h' \\ (x^3f'')' & (x^3g'')' & (x^3h'')' \end{vmatrix}$

55. Prove $\begin{vmatrix} yz - x^2 & zx - y^2 & xy - z^2 \\ zx - y^2 & xy - z^2 & yz - x^2 \\ xy - z^2 & yz - x^2 & zx - y^2 \end{vmatrix} =$

$$\begin{vmatrix} r^2 & u^2 & u^2 \\ u^2 & r^2 & u^2 \\ u^2 & u^2 & r^2 \end{vmatrix} \text{ where } r^2 = x^2 + y^2 + z^2$$

$$\text{and } u^2 = yz + zx + xy$$

56. Show that

$$\begin{vmatrix} a^2 + x^2 & ab - cx & ac + bx \\ ab + cx & b^2 + x^2 & bc + ax \\ ac - bx & bc - ax & c^2 + x^2 \end{vmatrix} = \begin{vmatrix} x & c & -b \\ -c & x & a \\ b & -a & x \end{vmatrix}^2$$

57. If a, b, c are all different and the points

$$\left(\frac{r^3}{r-1}, \frac{r^2-3}{r-1} \right) \text{ (where } r = a, b, c) \text{ are}$$

collinear then prove that

$$3(a+b+c) = ab+bc+ca-abc$$

KEY LEVEL-VI

SINGLE ANSWER QUESTIONS

- 1.D 2.B 3.B 4.C
5.B 6.A 7.B 8.B
9.B 10.C 11.C 12.A
13.C 14.A 15.A

MULTIPLE ANSWER QUESTIONS

- 16.ABC 17.ABCD 18.ABD

MATRIX MATCHING QUESTIONS

19. A - r, B - s, C - p, D - q,
20. A - r, B - r, C - q, D - s

COMPREHENSION QUESTIONS

- 21.C 22.A 23.B 24.D
25.C 26.D 27.A 28.B
29.C 30.A 31.D 32.A
33.C 34.A 35.D

INTEGER ANSWER QUESTIONS

- 36.4 37.1 38.6 39.7
40.0 41.0 42.3 43.0
44.2 45.1 46.2 47.1
48.4

HINTS LEVEL-VI

SINGLE ANSWER QUESTIONS

1. LHS = $\begin{vmatrix} \operatorname{cosec} \alpha & 1 & 0 \\ 1 & 2 \operatorname{cosec} \alpha & 1 \\ 0 & 1 & 2 \operatorname{cosec} \alpha \end{vmatrix}$

$$\begin{aligned}
 &= \begin{vmatrix} \operatorname{cosec} \alpha & 1 & 0 \\ 0 & 2 \operatorname{cosec} \alpha - \sin \alpha & 1 \\ 0 & 1 & 2 \operatorname{cosec} \alpha \end{vmatrix} [R_2 \rightarrow R_2 - \sin \alpha R_1] \\
 &= \operatorname{cosec} \alpha \begin{vmatrix} 2 \operatorname{cosec} \alpha - \sin \alpha & 1 \\ 1 & 2 \operatorname{cosec} \alpha \end{vmatrix} \\
 &= \operatorname{cosec} \alpha (4 \operatorname{cosec}^2 \alpha - 2 - 1) = 4 \operatorname{cosec}^3 \alpha - 3 \operatorname{cosec} \alpha
 \end{aligned}$$

$$\text{RHS} = \frac{1}{2} \left(\tan^3 \frac{\alpha}{2} + \cot^3 \frac{\alpha}{2} \right)$$

$$= \frac{1}{2} \left[\left(\tan \frac{\alpha}{2} + \cot \frac{\alpha}{2} \right)^3 - 3 \tan \frac{\alpha}{2} \cot \frac{\alpha}{2} \left(\tan \frac{\alpha}{2} + \cot \frac{\alpha}{2} \right) \right]$$

$$= \frac{1}{2} \left(\tan \frac{\alpha}{2} + \cot \frac{\alpha}{2} \right) \left[\left(\tan \frac{\alpha}{2} + \cot \frac{\alpha}{2} \right)^2 - 3 \right]$$

$$= \frac{1}{2} \frac{\sec^2 \alpha / 2}{\tan \alpha / 2} \left[\left(\frac{\sec^2 \alpha / 2}{\tan \alpha / 2} \right)^2 - 3 \right] = \frac{1}{2} \left(\frac{2}{\sin \alpha} \right) \left[\left(\frac{2}{\sin \alpha} \right)^2 - 3 \right]$$

$$= \operatorname{cosec} \alpha (4 \operatorname{cosec}^2 \alpha - 3)$$

$$= 4 \operatorname{cosec}^3 \alpha - 3 \operatorname{cosec} \alpha$$

$$2. \quad \text{LHS} = \begin{vmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix} \times \begin{vmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix}$$

$$= \frac{1}{abc} \begin{vmatrix} ax_1 & by_1 & cz_1 \\ ax_2 & by_2 & cz_2 \\ ax_3 & by_3 & cz_3 \end{vmatrix} \times \begin{vmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix}$$

$$= \frac{1}{abc} \begin{vmatrix} ax_1^2 + by_1^2 + cz_1^2 & ax_1x_2 + by_1y_2 + cz_1z_2 & ax_1x_3 + by_1y_3 + cz_1z_3 \\ ax_2x_1 + by_2y_1 + cz_2z_1 & ax_2^2 + by_2^2 + cz_2^2 & ax_2x_3 + by_2y_3 + cz_2z_3 \\ ax_3x_1 + by_3y_1 + cz_3z_1 & ax_3x_2 + by_3y_2 + cz_3z_2 & ax_3^2 + by_3^2 + cz_3^2 \end{vmatrix}$$

$$= \frac{1}{abc} \begin{vmatrix} \lambda & \mu & \mu \\ \mu & \lambda & \mu \\ \mu & \mu & \lambda \end{vmatrix} = \frac{1}{abc} (\lambda - \mu)^2 (\lambda + 2\mu)$$

$$3. \quad \Delta'(x) = \begin{vmatrix} 1 & \theta + x & \lambda + x \\ 1 & \phi + x & \mu + x \\ 1 & \psi + x & v + x \end{vmatrix}$$

$$+ \begin{vmatrix} \alpha + x & 1 & \lambda + x \\ \beta + x & 1 & \mu + x \\ \gamma + x & 1 & v + x \end{vmatrix} + \begin{vmatrix} \alpha + x & \theta + x & 1 \\ \beta + x & \phi + x & 1 \\ \gamma + x & \psi + x & 1 \end{vmatrix}$$

$$\Delta''(x) = 0$$

$$4. \quad \Delta_1 = \begin{vmatrix} \log 1 & \log \left(\frac{b}{c} \right) & \log \left(\frac{c}{a} \right) \\ \log 1 & \log \left(\frac{c}{a} \right) & \log \left(\frac{a}{b} \right) \\ \log 1 & \log \left(\frac{a}{b} \right) & \log \left(\frac{b}{c} \right) \end{vmatrix} c_1 \rightarrow c_1 \rightarrow c_2 + c_3 = 0$$

$$\text{Whereas } \Delta_2 = (3abc - a^3 - b^3 - c^3) \neq 0 \quad \frac{\Delta_1}{\Delta_2} = 0$$

$$5. \quad \sum_{k=0}^{n-1} 3^k = 1 + 3 + 3^2 + \dots + 3^{n-1} = \frac{3^n - 1}{3 - 1} = \frac{3^n - 1}{2}$$

$$\sum_{k=0}^{n-1} \frac{1}{(k+1)(k+2)} = \sum_{k=0}^{n-1} \left(\frac{1}{k+1} - \frac{1}{k+2} \right) = \frac{n}{n+1}$$

$$\sum_{k=0}^{n-1} \cos(k+1)\theta = 1 + \cos \theta + \dots + \cos(n-1)\theta$$

$$= \operatorname{Re} [1 + e^{i\theta} + e^{i2\theta} + \dots + e^{i(n-1)\theta}]$$

$$= \operatorname{Re} \left(\frac{1 - e^{in\theta}}{1 - e^{i\theta}} \right) = \operatorname{Re} \left(\frac{(1 - e^{in\theta})(1 - e^{-i\theta})}{(1 - e^{i\theta})(1 - e^{-i\theta})} \right)$$

$$= \operatorname{Re} \left[\frac{1 - e^{-i\theta} - e^{in\theta} + e^{i(n-1)\theta}}{2 - (e^{i\theta} + e^{-i\theta})} \right] = \frac{1 - \cos \theta - \cos \theta + \cos(n-1)\theta}{2(1 - \cos \theta)}$$

$$= \frac{2 \sin^2 \frac{\theta}{2} + 2 \sin \left(n - \frac{1}{2} \right) \theta \sin \frac{\theta}{2}}{4 \sin^2 \frac{\theta}{2}} = \frac{\sin \frac{\theta}{2} + \sin \left(n - \frac{1}{2} \right) \theta}{2 \sin \frac{\theta}{2}}$$

$$= \frac{\sin \frac{n\theta}{2} \cos \frac{(n-1)\theta}{2}}{\sin \frac{\theta}{2}}$$

$$6. \quad \Delta_C = \Delta^{4-1} = 3^3 = 27$$

$$7. \quad \text{Applying}$$

$$\begin{aligned} C_1 &\rightarrow C_1 - C_2 \\ C_2 &\rightarrow C_2 - C_3 \end{aligned}$$

$$\begin{aligned}
 \Delta &= \begin{vmatrix} 0 & 0 & 1 \\ \cot \frac{A}{2} - \cot \frac{B}{2} & \cot \frac{B}{2} - \cot \frac{C}{2} & \cot \frac{C}{2} \\ \tan \frac{B}{2} - \tan \frac{A}{2} & \tan \frac{C}{2} - \tan \frac{B}{2} & \tan \frac{A}{2} + \tan \frac{B}{2} \end{vmatrix} = 0 \\
 &= \left(\cot \frac{A}{2} - \cot \frac{B}{2} \right) \left(\cot \frac{B}{2} - \cot \frac{C}{2} \right)
 \end{aligned}$$

$$\times \begin{vmatrix} 0 & 0 & 1 \\ 1 & 1 & \cot \frac{C}{2} \\ \tan \frac{B}{2} \tan \frac{A}{2} & \tan \frac{B}{2} \tan \frac{C}{2} & \frac{\tan A}{2} + \tan \frac{B}{2} \end{vmatrix}$$

$$= \left(\cot \frac{A}{2} - \cot \frac{B}{2} \right) \left(\cot \frac{B}{2} - \cot \frac{C}{2} \right) \tan \frac{B}{2} \left(\tan \frac{C}{2} - \tan \frac{A}{2} \right) = 0$$

$$\Rightarrow \Delta = 0$$

$$\Rightarrow \cot \frac{A}{2} = \cot \frac{B}{2} \text{ or } \cot \frac{B}{2} = \cot \frac{C}{2} \text{ or } \tan \frac{A}{2} = \tan \frac{C}{2}$$

$$\Rightarrow \text{Triangle is definitely isosceles.}$$

$$8. \Delta = \begin{vmatrix} 1 & 4 & 9 & 16 \\ 4 & 9 & 16 & 25 \\ 9 & 16 & 25 & 36 \\ 16 & 25 & 36 & 49 \end{vmatrix}$$

$$(R_3 \rightarrow R_3 - R_2, R_4 \rightarrow R_4 - R_3)$$

$$= 3 \begin{vmatrix} 1 & 4 & 9 & 16 \\ 4 & 9 & 16 & 25 \\ 5 & 7 & 9 & 11 \\ 5 & 7 & 9 & 11 \end{vmatrix} \quad (R_4 \rightarrow R_4 - R_3)$$

9. We divide L.H.S by λ^4 and C_1 by λ^2 , C_2 by λ and C_3 by λ on the R.H.S to obtain

$$p + q \left(\frac{1}{\lambda} \right) + r \left(\frac{1}{\lambda} \right)^2 + s \left(\frac{1}{\lambda} \right)^3 + t \left(\frac{1}{\lambda} \right)^4$$

Taking limit as $\lambda \rightarrow \infty$, we get $p = -4$

10. We observe that the elements in the pre-factor are the cofactors of the corresponding elements of the post-factor. hence,

$$\begin{vmatrix} \lambda & c & -b \\ -c & \lambda & a \\ b & -a & \lambda \end{vmatrix} = \left[\lambda (\lambda^2 + a^2 + b^2 + c^2) \right]^3$$

$$= (1 + a^2 + b^2 + c^2)^3 \Rightarrow \lambda = 1$$

11. The given system is consistent

$$\therefore \Delta = \begin{vmatrix} 1 & 1 & -1 \\ 2 & -1 & -c \\ -b & 3b & -c \end{vmatrix} = 0 \Rightarrow c + bc - 6b + b + 2c + 3bc = 0$$

$$\Rightarrow c = \frac{5b}{4b+3}$$

Now $c < 1$

$$\Rightarrow \frac{5b}{4b+3} - 1 < 0 \Rightarrow b \in \left(-\frac{3}{4}, 3 \right)$$

12.

$$g(x) = \begin{vmatrix} a^{-x} & e^{\log_e a^x} & x^2 \\ a^{-3x} & e^{\log_e a^{3x}} & x^4 \\ a^{-5x} & e^{\log_e a^{5x}} & 1 \end{vmatrix} = \begin{vmatrix} a^{-x} & a^x & x^2 \\ a^{-3x} & a^{3x} & x^4 \\ a^{-5x} & a^{5x} & 1 \end{vmatrix} \quad (e^{\log_e a^x} = a^x)$$

$$= g(-x) = \begin{vmatrix} a^{-(-x)} & a^{-x} & (-x)^2 \\ a^{-3(-x)} & a^{-3x} & (-x)^4 \\ a^{-5(-x)} & a^{-5x} & 1 \end{vmatrix} = - \begin{vmatrix} a^{-x} & a^x & x^2 \\ a^{-3x} & a^{3x} & x^4 \\ a^{-5x} & a^{5x} & 1 \end{vmatrix}$$

$$= -g(x)$$

$$= g(x) = g(-x) = 0$$

$= g(x)$ is an odd function

13. In the left hand determinant, each element is the cofactor of the elements of the determinant

$$\begin{vmatrix} x & y & z \\ y & z & x \\ z & x & y \end{vmatrix} = \Delta^* \text{ (say)}$$

$$\text{Hence } \Delta^{*2} = \begin{vmatrix} x & y & z \\ y & z & x \\ z & x & y \end{vmatrix} \begin{vmatrix} x & y & z \\ y & z & x \\ z & x & y \end{vmatrix} = \begin{vmatrix} r^2 & u^2 & u^2 \\ u^2 & r^2 & u^2 \\ u^2 & u^2 & r^2 \end{vmatrix}$$

$$14. \Delta = \begin{vmatrix} x & y & z \\ p & q & r \\ a & b & c \end{vmatrix} \begin{vmatrix} 0 & m & n \\ -m & 0 & k \\ -n & -k & 0 \end{vmatrix} \text{ where } \begin{vmatrix} 0 & m & n \\ -m & 0 & k \\ -n & -k & 0 \end{vmatrix}$$

is skew symmetric.

$$\therefore \Delta = 0$$

15. Given, system of equations is

$$x - 2y + 3z = -1$$

$$x - 3y + 4z = 1$$

$$-x + y - 2z = k$$

$$\therefore D = \begin{vmatrix} 1 & -2 & 3 \\ 1 & -3 & 4 \\ -1 & 1 & -2 \end{vmatrix}$$

$$= 1(6-4) + 2(-2+4) + 3(1-3)$$

DETERMINANTS

$$= 2 + 4 - 6 = 0$$

As per given determinant in statement II, we find

D_2

$$D_2 = \begin{vmatrix} 1 & -1 & 3 \\ 1 & 1 & 4 \\ -1 & k & -2 \end{vmatrix} - \begin{vmatrix} 1 & 3 & -1 \\ 1 & 4 & 1 \\ -1 & -2 & k \end{vmatrix}$$

$$C_2 \leftrightarrow C_3$$

$$= \begin{vmatrix} 1 & 3 & -1 \\ -1 & -2 & k \\ 1 & 4 & 1 \end{vmatrix}$$

$$R_2 \leftrightarrow R_3$$

$$= 1(-2 - 4k) - 3(-1 - k) - 1(-4 + 2)$$

$$= 3 - k$$

$$\therefore D_2 \neq 0 \text{ for } k \neq 3$$

$\Rightarrow$ statement II is true

And also for $k \neq 3, D_2 \neq 0$ but $D = 0$

$\therefore$ the system of equations has no solution

$\Rightarrow$ statement I is true and justified by statement II

MULTIPLE ANSWER QUESTIONS

16. Applying

$$R_3 \rightarrow R_3 - R_2, \Delta = \begin{vmatrix} x^2 - a^2 & x^2 - b^2 & x^2 - c^2 \\ (x-a)^3 & (x-b)^3 & (x-c)^3 \\ 6x^2a + 2a^3 & 6x^2b + 2b^3 & 6x^2c + 2c^3 \end{vmatrix}$$

$$R_3 \rightarrow \frac{1}{2}R_3 \text{ then } R_2 \rightarrow R_2 + R_3, \text{ we get}$$

$$\Delta = 2 \begin{vmatrix} x^2 - a^2 & x^2 - b^2 & x^2 - c^2 \\ x^3 + 3xa^2 & x^3 + 3xb^2 & x^3 + 3xc^2 \\ 3x^2a + a^3 & 3x^2b + b^3 & 3x^2c + c^3 \end{vmatrix}$$

$$R_2 \rightarrow \left(\frac{1}{x}\right)R_2 \text{ \& then } C_2 \rightarrow C_2 - C_1, C_3 \rightarrow C_3 - C_1$$

$$\Delta = 2x \begin{vmatrix} x^2 - a^2 & a^2 - b^2 & a^2 - c^2 \\ x^2 + 3a^2 & 3(b^2 - a^2) & 3(c^2 - a^2) \\ 3x^2a + a^3 & 3x^2(b-a) + (b^3 - a^3) & 3x^2(c-a) + (c^3 - a^3) \end{vmatrix}$$

$$= 2x(b-a)(c-a) \begin{vmatrix} x^2 - a^2 & -(a+b) & -(a+c) \\ x^2 + 3a^2 & 3(b+a) & 3(c+a) \\ 3x^2a + a^3 & 3x^2 + b^2 + ab + a^2 & 3x^2 + c^2 + a^2 + ac \end{vmatrix}$$

Applying $R_2 \rightarrow R_2 + 3R_1$ & $R_3 \rightarrow R_3 + aR_1$

$$\Delta = 2x(b-a)(c-a) \begin{vmatrix} x^2 - a^2 & -(a+b) & -(a+c) \\ 4x^2 & 0 & 0 \\ 4x^2a & 3x^2 + b^2 & 3x^2 + c^2 \end{vmatrix}$$

$$= 2x(b-a)(c-a)4x^2[(a+b)(3x^2 + c^2) - (a+c)(3x^2 + b^2)]$$

$$= 8x^3(b-a)(c-a)(b-c)\{3x^2 - (bc + ca + ab)\}$$

As a, b, c are distincts $\Delta = 0$

$$x = 0 \text{ or } x^2 = \frac{1}{3}(bc + ca + ab)$$

$$\Rightarrow x = 0, \pm \sqrt{\frac{1}{3}(bc + ca + ab)}$$

$$17. \Delta = \frac{1}{abc} \begin{vmatrix} a^3 + ax^2 & a^2b & a^2c \\ ab^2 & b^3 + x^2b & b^2c \\ ac^2 & bc^2 & c^3 + cx^2 \end{vmatrix}$$

$$= \frac{1}{abc} \times abc \begin{vmatrix} a^2 + x^2 & a^2 & a^2 \\ b^2 & b^2 + x^2 & b^2 \\ c^2 & c^2 & c^2 + x^2 \end{vmatrix}$$

$$R_1 \rightarrow R_1 + R_2 + R_3$$

$$= (a^2 + b^2 + c^2 + x^2) \begin{vmatrix} 1 & 1 & 1 \\ b^2 & b^2 + x^2 & b^2 \\ c^2 & c^2 & c^2 + x^2 \end{vmatrix}$$

Applying $C_2 \rightarrow C_2 - C_1, C_3 \rightarrow C_3 - C_1$, and on solving

$$= (a^2 + b^2 + c^2 + x^2)x^4$$

18.

$$\Delta = (abc)^3 \begin{vmatrix} \frac{1}{a} & \frac{1}{b} & \frac{1}{c} \\ \frac{1}{b} & \frac{1}{c} & \frac{1}{a} \\ \frac{1}{c} & \frac{1}{a} & \frac{1}{b} \end{vmatrix} = (abc)^3 \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right) \begin{vmatrix} 1 & 1 & 1 \\ \frac{1}{b} & \frac{1}{c} & \frac{1}{a} \\ \frac{1}{c} & \frac{1}{a} & \frac{1}{b} \end{vmatrix}$$

On solving

$$\Delta = abc^3 \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right) \left(\frac{1}{bc} + \frac{1}{ca} + \frac{1}{ab} - \frac{1}{a^2} - \frac{1}{b^2} - \frac{1}{c^2} \right)$$

$$\Delta = abc^3 \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right) \left(\frac{1}{a} + \frac{w}{b} + \frac{w^2}{c} \right) \left(\frac{1}{a} + \frac{w^2}{b} + \frac{w}{c} \right)$$

MATRIX MATCHING QUESTIONS

19. (A) Since every entry can take 2 values.

Total passing determinant $= 2^9 = 512$.

(B) The minimum value of trace is -3 and maximum value is 3 hence sum of them = 0.

(C) The maximum value of such a determinant is 4 attained

$$\text{by } \begin{vmatrix} 1 & -1 & 1 \\ 1 & 1 & -1 \\ -1 & 1 & 1 \end{vmatrix}.$$

(D) The matrix trace is obviously three.

20. A.

$$f'(x) = \begin{vmatrix} -\sin x & 1 & 0 \\ 2\sin x & x^2 & 2x \\ \tan x & x & 1 \end{vmatrix} + \begin{vmatrix} \cos x & x & 1 \\ 2\cos x & 2x & 2 \\ \tan x & x & 1 \end{vmatrix} + \begin{vmatrix} \cos x & x & 1 \\ 2\sin x & x^2 & 2x \\ \sec^2 x & 1 & 0 \end{vmatrix}$$

B. $\int_0^{\pi/2} f'(x) dx = f\left(\frac{\pi}{2}\right) - f(0)$

C. 2

$$f(x)f\left(\frac{1}{x}\right) = f(x) + f\left(\frac{1}{x}\right)$$

Comparing coefficient of x^n and constant term

$$ab = b \Rightarrow a = 1$$

$$a^2 + b^2 = 2a$$

$$1 + b^2 = 2 \Rightarrow b = \pm 1$$

$$f(x) = 1 \pm x^n$$

$$f(2) = 17 = 1 + 2^n \Rightarrow x = 4$$

$$f(5) = 1 + 5^4 = 626$$

D. $k = 1$

$$C_1 \rightarrow C_1 - 2\sin x C_3$$

$$C_2 \rightarrow C_2 + 2\cos x C_3$$

$$\text{On solving } \Delta = 2 \text{ and } \Delta' = 0$$

$$\therefore \int_0^{\pi/2} (\Delta + \Delta') dx = \pi$$

COMPREHENSION QUESTIONS

Passage-I

21,22,23

$$R_2 \rightarrow R_2 - 20R_1$$

$$R_3 \rightarrow R_3 - 2R_1$$

24. Given $A = \begin{bmatrix} a & b \\ c & a \end{bmatrix}$, $a, b, c \in \{0, 1, 2, \dots, (p-1)\}$

If A is skew-symmetric matrix, then $a=0, b=-c$

$$\therefore |A| = -b^2$$

Thus, p divides $|A|$ only when $b=0$

.....(i)

Again, If A is symmetric matrix, then $b=c$ and

$$|A| = a^2 - b^2$$

Thus, p divides $|A|$ if either p divides $(a-b)$ or p divides $(a+b)$. p divides only when $a=b$

$$\text{i.e. } a=b \in \{0, 1, 2, \dots, (p-1)\}$$

i.e p choices

.....(ii)

p divides $(a+b)$

$\Rightarrow$ p choices, including $a=b=0$ included in Eq.

(i)

$\therefore$ total number of choices are $(p + p - 1) = 2p - 1$

Hence (d) is the correct option

25. Trace of $A = 2a$, will be divisible by p if $a=0$

$$|A| = a^2 - bc, \text{ for } (a^2 - bc) \text{ to be divisible by p.}$$

There are exactly $(p-1)$ ordered pairs (b, c) for any value of a.

$\therefore$ required number is $(p-1)^2$

Hence, (c) is the correct option

26. The number of matrices for which p does not divide $\text{Tr}(A) = (p-1)p^2$ of these $(p-1)^2$ are such that p divides $|A|$. The number of matrices for which p divides $\text{Tr}(A)$ and p does not divides $|A|$ are $(p-1)^2$

27. $abc(ab + bc + ca) = a + b + c$

$$\Rightarrow a\left(\frac{3-a^2c^2}{2}\right) + c\left(\frac{3-a^2c^2}{2}\right) + a^2bc^2 = a + b + c$$

$$\Rightarrow (2b - (a+c))(a^2c^2 - 1) = 0$$

$$\Rightarrow a + c = 2b$$

28. $b^4 = 1 \Rightarrow b = 1$

$$ac(a + c + ac) = a + c + 1$$

$$\Rightarrow (ac - 1)(a + 1)(c + 1) = 0$$

$$a = -1, c = -1$$

$$\Rightarrow ac = b^2$$

29. $a + b + c = 3a^2bc^2$

$$\Rightarrow abc(ab + bc + ca) = 3a^2bc^2$$

$$\Rightarrow \frac{1}{c} + \frac{1}{a} = \frac{2}{b}$$

30,31,32.

$$\Delta = \begin{vmatrix} 1 & 1 & 1 \\ 1 & \alpha & \beta \\ 1 & \alpha^2 & \beta^2 \end{vmatrix} \begin{vmatrix} 1 & 1 & 1 \\ \alpha & \alpha^2 \\ \beta & \beta^2 \end{vmatrix} = (\alpha - \beta)^2 (\alpha + \beta - 1 - \alpha\beta)^2$$

$$\Delta = \begin{vmatrix} -1 & c & b \\ c & -1 & a \\ b & a & -1 \end{vmatrix} = 0$$

$$= a^2 + b^2 + c^2 = 2abc = 1$$

33,34,35

$$\frac{x}{ac + b} = \frac{y}{bc + a} = \frac{z}{1 - c^2}$$

$$\frac{x^2}{(ac+b)^2} = \frac{y^2}{(bc+a)^2} = \frac{z^2}{(1-c^2)^2} \dots(1)$$

$$\frac{x^2}{1-a^2} = \frac{y^2}{1-b^2} = \frac{z^2}{1-c^2} \dots(2)$$

From (1), we that $1-a^2, 1-b^2, 1-c^2$ are all positive or all negative. Given that one of a, b, c is proper fraction, so

$$1-a^2 > 0, 1-b^2 > 0, 1-c^2 > 0, \text{ which gives}$$

$$a^2 + b^2 + c^2 < 3 \dots(3)$$

Using (1) and (3), we get

$$1 < 3 + 2abc$$

or

$$abc > -1 \dots(4)$$

INTEGER ANSWER QUESTIONS

36. Since on expanding the determinant it will become a symmetric homogeneous function of degree 3 in a, b, c and R.H.S. is also so (k is a numerical quantity) k can be obtained by assigning numerical values of a, b, c .

On putting $a = b = c = 1$

$$\text{We get } \begin{vmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{vmatrix} = kabc$$

$$\Rightarrow 4 = kabc$$

$$\Rightarrow k = 4$$

$$37. D = \begin{vmatrix} a & 1 & 1 \\ 1 & b & 1 \\ 1 & 1 & c \end{vmatrix} = abc = a + b + c + 2$$

$$D_1 = \begin{vmatrix} a & 1 & 1 \\ b & b & 1 \\ c & 1 & c \end{vmatrix} = abc = a + b + c = 2bc$$

$$D_2 = \begin{vmatrix} a & a & 1 \\ 1 & b & 1 \\ 1 & c & c \end{vmatrix} = abc = b + a + c = 2ab$$

$$D_3 = \begin{vmatrix} a & 1 & a \\ 1 & b & b \\ 1 & 1 & c \end{vmatrix} = abc = c + a + b = 2ab$$

For the system of equation has no solution $D = 0$ and atleast one D_1, D_2, D_3 is non-zero.

$$\text{Hence } D = abc - a - b - c + 2 = 0$$

Now, D_1, D_2 and D_3 becomes

$$D_1 = abc - a - b + 2b + 2c = 2bc - 2$$

$$= 0 + 2b + 2c - 2bc = 2$$

$$= -2(1-b)(1-b)$$

Therefore D_1, D_2, D_3 cannot vanish if a, b, c are different from 1.

Hence the number is 1.

38. For non-trivial solution

$$D = \begin{vmatrix} \sin 3\theta & 1 & 1 \\ \cos 2\theta & 4 & 3 \\ 2 & 7 & 7 \end{vmatrix} = 0$$

$$\Rightarrow \sin 3\theta + 2\cos 2\theta = 2$$

$$\Rightarrow 4\sin^3\theta + 4\sin^2\theta = 3\sin\theta$$

$$\Rightarrow \sin\theta = 0 \text{ or } \sin\theta = \frac{1}{2} \left(\because \sin\theta \neq \frac{-3}{2} \right)$$

$$\Rightarrow \sin\theta = \sin\frac{\pi}{6}$$

$$\Rightarrow \theta = \frac{5\pi}{6}$$

$$\Rightarrow \lambda = 6$$

$$39. \sum_{i=1}^i D_i = D_1 + D_2 + D_3 + \dots + D_n = 56$$

$$\begin{vmatrix} 1 & n & n \\ 2 & n^2 + n + 1 & n^2 + n \\ 1 & n^2 & n^2 + n + 1 \end{vmatrix} + \begin{vmatrix} 1 & n & n \\ 4 & n^2 + n + 1 & n^2 + n \\ 3 & n^2 & n^2 + n + 1 \end{vmatrix}$$

$$+ \dots + \begin{vmatrix} 1 & n & n \\ 2n & n^2 + n + 1 & n^2 + n \\ 2n-1 & n^2 & n^2 + n + 1 \end{vmatrix} = 56$$

$$\begin{vmatrix} 1+1+\dots+1 & n & n \\ 2+4+\dots+2n & n^2 + n + 1 & n^2 + n \\ 1+3+5+\dots+2n-1 & n^2 & n^2 + n + 1 \end{vmatrix} = 56$$

$$\begin{vmatrix} n & n & n \\ n(n+1) & n^2 + n + 1 & n^2 + n \\ n^2 & n^2 & n^2 + n + 1 \end{vmatrix} = 56$$

$$C_1 \rightarrow C_2 - C_1, C_3 - C_1$$

$$\begin{vmatrix} 1 & 0 & 0 \\ n & n^2 + n & 1 \\ n^2 & 0 & n + 1 \end{vmatrix} = 56$$

$$n(n+1) = 56$$

$$\Rightarrow n = -8, 7 \text{ but } n \neq -8$$

$$\Rightarrow n = 7$$

40. $f(x)$ = determinant of $\Delta(x)$

$$= -2(x+4)(x-1)^2$$

$$f'(x) = 0 \Rightarrow x = 1, \frac{7}{3}$$

$$f(-3) = -ve, f(1) = 0$$

$$f\left(\frac{7}{3}\right) \text{ is negative, } f(18) \text{ is negative}$$

$$\Rightarrow \text{maximum of } f(x) \text{ is } 0$$

$$41. \text{ Use } \sum \tan A = \prod \tan A$$

$$42. \text{ multiply } R_1, R_2, R_3 \text{ by } x, y, z \text{ respectively, and take common, } x, y, z \text{ from } C_1, C_2, C_3 \text{ respectively then determinant becomes,}$$

$$(x^3 + y^3 + z^3 + 1) \begin{vmatrix} 0 & 0 & 1 \\ -1 & 1 & y^3 \\ 0 & -1 & z^3 + 1 \end{vmatrix} = 11$$

$$\Rightarrow x^3 + y^3 + z^3 = 10$$

$$\Rightarrow x = 1, y = 1, z = 2 \quad (\text{or}) \quad x = 1, y = 2, z = 1 \quad (\text{or})$$

$$x = 2, y = 1, z = 1$$

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$$\Rightarrow a_k = a_{16-k}, 0 \leq k \leq 16$$

$$\therefore \begin{vmatrix} a_3 & a_7 & a_{13} \\ a_4 & a_6 & a_{12} \\ a_5 & a_4 & a_{11} \end{vmatrix} = \begin{vmatrix} a_3 & a_7 & a_3 \\ a_4 & a_6 & a_4 \\ a_5 & a_4 & a_5 \end{vmatrix} = 0$$

$$44. \text{ We have } D = \begin{vmatrix} a_1 & a_2 & a_3 \\ 5 & 4 & a_6 \\ a_7 & a_8 & a_9 \end{vmatrix}$$

$$\text{Since } a_n = \frac{20}{n}; d = \frac{1}{20}$$

$$\text{Hence } R_1 \rightarrow R_1 - R_2 \text{ and } R_2 \rightarrow R_2 - R_3$$

$$= \frac{(20)^3}{4 \times 7} \begin{vmatrix} 0 & \frac{-3}{10} & \frac{-1}{3} \\ 0 & \frac{-3}{40} & \frac{-1}{9} \\ 1 & \frac{7}{8} & \frac{7}{9} \end{vmatrix} = \frac{50}{21} \Rightarrow [D] = 2$$

$$45. [C_3 \rightarrow C_3 - C_2]$$

$$\Delta_1 = -2d^4 \text{ Similarly } \Delta_2 = -2d^4$$

$$46. \text{ Let}$$

$$a = \begin{vmatrix} 2^2 a_{11} & 2^3 a_{21} & 2^4 a_{31} \\ 2^3 a_{21} & 2^4 a_{22} & 2^5 a_{32} \\ 2^4 a_{31} & 2^5 a_{32} & 2^6 a_{33} \end{vmatrix} = 2^9 \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ 2a_{21} & 2a_{22} & 2a_{33} \\ 2^2 a_{31} & 2^2 a_{32} & 2^2 a_{33} \end{vmatrix}$$

$$2^{12} \det P = 2^{13}$$

$$47. \text{ Let } A = \begin{bmatrix} 1 & \omega & \omega^2 \\ \omega & \omega^2 & 1 \\ \omega^2 & 1 & \omega \end{bmatrix} \text{ Now, } A^2 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\text{And } \text{Tr}(A) = 0, |A| = 0$$

$$\therefore A^3 = 0$$

$$\Rightarrow \begin{vmatrix} z+1 & \omega & \omega^2 \\ \omega & z+\omega^2 & 1 \\ \omega & 1 & z+\omega \end{vmatrix} = |A + zI| = 0 \Rightarrow z^3 = 0$$

$$\Rightarrow z = 0, \text{ then number of } z \text{ satisfying the given equation is } 1.$$

$$48. |A| = (2k+1)^3, |B| = 0$$

$$\text{But } \det(\text{adj} A) + \det(\text{adj} B) = 10^6$$

$$\Rightarrow (2k+1)^6 = 10^6 \Rightarrow k = \frac{9}{2} \Rightarrow [k] = 4$$

SUBJECTIVE QUESTIONS

$$49. \text{ Put } (m+n-l-p)^2 = A \text{ etc.}$$

$$50. \Delta = -(a^3 + b^3 + c^3 - 3abc) = 0$$

$$\therefore a^3 + b^3 + c^3 - 3abc = 0$$

$$\frac{1}{2}(a+b+c)[(a-b)^2 + (b-c)^2 + (c-a)^2] = 0$$

$$\text{Above relation implies that either } a+b+c=0$$

$$\text{Or } a = b = c$$

$$\text{If } a+b+c=0, \text{ then we must have}$$

$$\cos \theta + \cos 3\theta + \cos 2\theta = 0$$

$$\sin \theta + \sin 3\theta - \sin 2\theta = 0$$

$$\text{Or } \cos 2\theta(2 \cos \theta + 1) = 0$$

$$\text{And } \sin 2\theta(2 \cos \theta - 1) = 0$$

$$\text{The above equations do not hold simultaneously because } \cos 2\theta = 0 \text{ i.e., } \theta = 45^\circ \text{ then second equation is not satisfied and if } 2 \cos \theta + 1 = 0$$

$$\text{Or } \cos \theta = \frac{1}{2} \text{ i.e., } \theta = 120^\circ \text{ then also}$$

$$\text{second equation is not satisfied.}$$

$$\text{Therefore the only possibility is } a = b = c.$$

$$\text{Or } e^{i\theta} = e^{-2i\theta} = e^{3i\theta}$$

$$\text{Above is satisfied only when } e^{i\theta} = 1.$$

$$\text{i.e. } \cos \theta + i \sin \theta = 1$$

$$\therefore \cos \theta = 1 \text{ and } \sin \theta = 0 \quad \therefore \theta = 2n\pi.$$

$$51. \text{ Put } x = 0, \text{ then } \begin{vmatrix} 0 & -1 & 3 \\ 1 & 2 & -3 \\ -3 & 4 & 0 \end{vmatrix} = 21$$

$$\text{Now differentiate both sides and put } x = 0$$